

Question 1

The members of the truss shown in Figure 1 have different cross-sectional areas and are made of different Young's moduli. Determine the deflection of joint A.

The member BC: $A = 32.3 \text{ cm}^2$, $E = 6.9 \times 10^{10} \text{ N/m}^2$, $L = 2.54 \text{ m}$

The member AB: $A = 38.7 \text{ cm}^2$, $E = 20.7 \times 10^{10} \text{ N/m}^2$, $L = 2.54 \text{ m}$

The member AC: $A = 25.8 \text{ cm}^2$, $E = 20.7 \times 10^{10} \text{ N/m}^2$, $L = 3.59 \text{ m}$

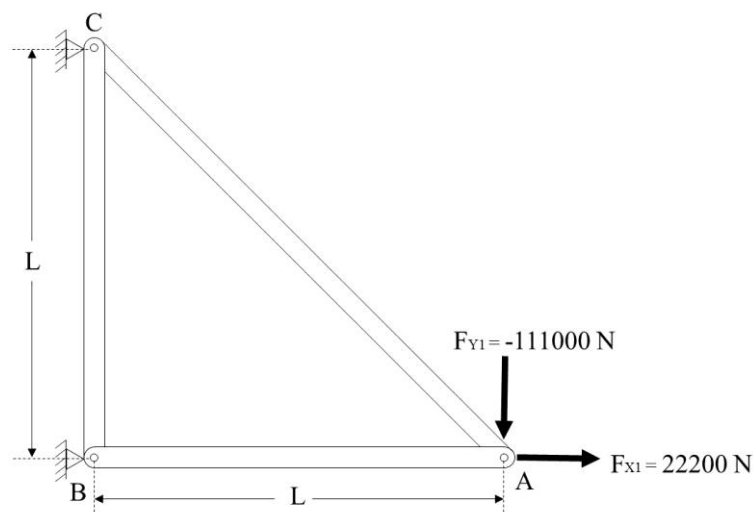


Figure 1

The system of SI units is chosen.



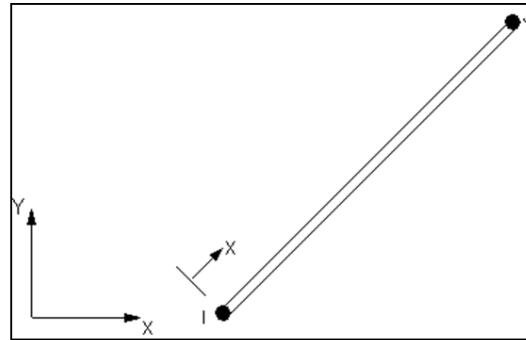
- Ansys had no build-in unit system
- The unit must be consistent.

Table 2-1 Consistent units.

Quantity	SI	SI (mm)	US Unit (ft)	US Unit (inch)
Length	m	mm	ft	in
Force	N	N	lbf	lbf
Mass	kg	tonne (10 ³ kg)	slug	lbf s ² /in
Time	s	s	s	s
Stress	Pa (N/m ²)	MPa (N/mm ²)	lbf/ft ²	psi (lbf/in ²)
Energy	J	mJ (10 ⁻³ J)	ft lbf	in lbf
Density	kg/m ³	tonne/mm ³	slug/ft ³	lbf s ² /in ⁴

LINK1

2-D spar



Element Name	LINK1
Nodes	I, J
Degrees of Freedom	UX, UY
Real Constants	AREA, ISTRN
Material Properties	EX, ALPX, DENS, DAMP
Surface Loads	None
Body Loads	Temperature -- T(I), T(J)
Special Features	Plasticity, Creep, Stress stiffening, Large deflection

/POST1

PLDISP,1

PRNSOL, U, COMP

FINISH

! displaced structure

! nodal solution of displacements